CASE STUDY 1

Unit I: Algorithmic Strategies

E-commerce Product Sorting and Google Calendar Scheduling

Aim

To understand Divide and Conquer through product sorting and the Greedy strategy through event scheduling.

Objectives

- Sort product prices using Merge Sort.
- Select non-overlapping events using a greedy approach.
- Understand the use of algorithms in everyday applications.

Theory

Divide and Conquer breaks a problem into smaller parts, solves them, and combines the results. Merge Sort uses this method to sort product prices. A Greedy algorithm makes a suitable choice at each step. In activity selection, events are sorted by finish time and an event is selected if it does not overlap with the last selected event.

Algorithm

1. Merge Sort: divide the list into halves, sort both halves recursively, and merge them in sorted order.

2. Greedy scheduling: sort events by finish time; select the first event; then select each event that starts after the previously selected event finishes.

Python Program

```
# Program 1: Sort product prices using Merge Sort
def merge_sort(prices):
    if len(prices) <= 1:
        return prices

    mid = len(prices) // 2
    left = merge_sort(prices[:mid])
    right = merge_sort(prices[mid:])
    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result

prices = [900, 250, 700, 400]
print("Sorted prices:", merge_sort(prices))


# Program 2: Select non-overlapping calendar events
events = [(9, 10, "A"), (9, 11, "B"), (10, 11, "C"), (11, 12, "D")]
events.sort(key=lambda event: event[1])

selected = []
last_finish = -1
for start, finish, name in events:
    if start >= last_finish:
        selected.append(name)
        last_finish = finish

print("Selected events:", selected)
```

Sample Output

```
Sorted prices: [250, 400, 700, 900]\nSelected events: ['A', 'C', 'D']
```

Conclusion

Merge Sort efficiently sorts product prices, while the greedy activity-selection method helps choose a set of non-overlapping events.